Exam Topic for Oral Presentation
EduQual Level 6

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Topic 78: Implementing Intelligent Infrastructure Monitoring Platform with AIOps, Predictive Maintenance, and Automated Remediation for Hybrid Cloud Environments

1. **Overview:**
   This project challenges students to architect and implement a comprehensive intelligent infrastructure monitoring platform supporting AIOps, predictive maintenance, and automated remediation for hybrid cloud environments. The solution must provide proactive monitoring, intelligent alerting, and automated problem resolution while optimizing infrastructure performance and availability. Students will build an AIOps platform combining machine learning, automation, and comprehensive observability.
2. **Key Learning Objectives:**
   a. **AIOps Implementation and Intelligent Monitoring:**
      i. Deploy machine learning models for infrastructure anomaly detection and pattern recognition
      ii. Implement root cause analysis automation using correlation algorithms and dependency mapping
      iii. Configure intelligent alerting with noise reduction and alert correlation
   b. **Predictive Maintenance and Capacity Planning:**
      i. Deploy predictive models for hardware failure prediction and maintenance scheduling
      ii. Implement capacity forecasting and resource optimization algorithms
      iii. Configure automated capacity planning and scaling recommendations
   c. **Automated Remediation and Self-Healing Systems:**
      i. Deploy automated remediation workflows for common infrastructure issues
      ii. Implement self-healing capabilities with rollback and safety mechanisms
      iii. Configure runbook automation and orchestrated response procedures
   d. **Hybrid Cloud Monitoring and Integration:**
      i. Deploy unified monitoring across on-premises, cloud, and edge infrastructure
      ii. Implement multi-cloud observability with consistent metrics and alerting
      iii. Configure hybrid network monitoring and performance optimization
   e. **Performance Analytics and Optimization:**
      i. Deploy application performance monitoring (APM) integration for full-stack visibility
      ii. Implement infrastructure performance analysis and bottleneck identification
      iii. Configure optimization recommendations and automated tuning
   f. **Compliance and SLA Management:**
      i. Deploy automated SLA monitoring and compliance reporting
      ii. Implement performance baseline establishment and drift detection

iii. Configure automated compliance validation and audit trail management

g. **Industry Application:** Used by enterprise IT operations teams, managed service providers, cloud platform operators, and DevOps organizations implementing intelligent infrastructure management and AIOps capabilities.

h. **Open Source and Cloud-Native Tools:**
   i. **Monitoring:** Prometheus, Grafana, Nagios, Zabbix
   ii. **AIOps/ML:** TensorFlow, scikit-learn, Apache Spark, MLflow
   iii. **Automation:** Ansible, Rundeck, Apache Airflow
   iv. **APM:** Jaeger, Zipkin, OpenTelemetry
   v. **Log Analytics:** ELK Stack, Loki, Fluentd
   vi. **Integration:** REST APIs, webhooks, message queues
   vii. **Cloud Platforms:** AWS CloudWatch, Azure Monitor, GCP Operations

3. **Examination Format:**
   a. **Presentation (15–20 minutes):**
      i. Intelligent infrastructure monitoring platform architecture with AIOps
      ii. Predictive maintenance and automated remediation demonstration
      iii. Hybrid cloud monitoring and performance analytics implementation
      iv. Compliance and SLA management automation showcase
   b. **Interview (30–40 minutes):**
      i. AIOps principles and intelligent monitoring strategies
      ii. Predictive maintenance and automated remediation challenges
      iii. Hybrid cloud monitoring and performance optimization techniques
      iv. Various questions will be asked as per the topic scope.